



Proportional or Not? Teaching Bundle

Grade 7 • North Carolina • Mathematics

NC.7.RP.2 • Identify and represent proportional relationships.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Identify and represent proportional relationships. This bundle practices a focused part of NC.7.RP.2; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Compare y/x at nonzero x values. A constant ratio gives $y = kx$ and a graph through the origin.

Example 1: A table has (x,y) pairs $[(1, 6), (2, 9), (4, 15)]$. Is the relationship proportional? Explain using ratios and write a rule.

Ratios y/x are 6, $9/2$, $15/4$. They differ. The starting value is not zero.

Example 2: A table has (x,y) pairs $[(1, 8), (2, 16), (4, 32)]$. Is the relationship proportional? Explain using ratios and write a rule.

Ratios y/x are 8, 8, 8. They are equal, so the constant of proportionality is 8.

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

A table has (x,y) pairs [(1, 5), (2, 8), (4, 14)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 02

A table has (x,y) pairs [(1, 4), (2, 8), (4, 16)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 03

A table has (x,y) pairs [(1, 5), (2, 8), (4, 14)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 04

A table has (x,y) pairs [(1, 9), (2, 18), (4, 36)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

A table has (x,y) pairs [(1, 12), (2, 19), (4, 33)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 06

A table has (x,y) pairs [(1, 5), (2, 10), (4, 20)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 07

A table has (x,y) pairs [(1, 9), (2, 13), (4, 21)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 08

A table has (x,y) pairs [(1, 2), (2, 4), (4, 8)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

A table has (x,y) pairs [(1, 8), (2, 10), (4, 14)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 10

A table has (x,y) pairs [(1, 4), (2, 8), (4, 16)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 11

A table has (x,y) pairs [(1, 13), (2, 20), (4, 34)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 12

A table has (x,y) pairs [(1, 6), (2, 12), (4, 24)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

A table has (x,y) pairs [(1, 11), (2, 16), (4, 26)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 14

A table has (x,y) pairs [(1, 3), (2, 6), (4, 12)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 15

A table has (x,y) pairs [(1, 10), (2, 13), (4, 19)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

CARD 16

A table has (x,y) pairs [(1, 4), (2, 8), (4, 16)]. Is the relationship proportional? Explain using ratios and write a rule.

Explain your method.

Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. A table has (x,y) pairs $[(1, 10), (2, 18), (4, 34)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 5), (2, 10), (4, 20)]$. Is the relationship proportional? Explain using ratios and write a rule.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. A table has (x,y) pairs $[(1, 9), (2, 12), (4, 18)]$. Is the relationship proportional? Explain using ratios and write a rule.

4. A table has (x,y) pairs $[(1, 7), (2, 14), (4, 28)]$. Is the relationship proportional? Explain using ratios and write a rule.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. A table has (x,y) pairs $[(1, 8), (2, 11), (4, 17)]$. Is the relationship proportional? Explain using ratios and write a rule.

6. A table has (x,y) pairs $[(1, 2), (2, 4), (4, 8)]$. Is the relationship proportional? Explain using ratios and write a rule.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. A table has (x,y) pairs $[(1, 10), (2, 13), (4, 19)]$. Is the relationship proportional? Explain using ratios and write a rule.

8. A table has (x,y) pairs $[(1, 3), (2, 6), (4, 12)]$. Is the relationship proportional? Explain using ratios and write a rule.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. A table has (x,y) pairs $[(1, 13), (2, 19), (4, 31)]$. Is the relationship proportional? Explain using ratios and write a rule.

10. A table has (x,y) pairs $[(1, 2), (2, 4), (4, 8)]$. Is the relationship proportional? Explain using ratios and write a rule.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A table has (x,y) pairs $[(1, 13), (2, 22), (4, 40)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 2), (2, 4), (4, 8)]$. Is the relationship proportional? Explain using ratios and write a rule.

3. A table has (x,y) pairs $[(1, 10), (2, 15), (4, 25)]$. Is the relationship proportional? Explain using ratios and write a rule.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A table has (x,y) pairs $[(1, 8), (2, 16), (4, 32)]$. Is the relationship proportional? Explain using ratios and write a rule.

5. A table has (x,y) pairs $[(1, 6), (2, 10), (4, 18)]$. Is the relationship proportional? Explain using ratios and write a rule.

6. A table has (x,y) pairs $[(1, 6), (2, 12), (4, 24)]$. Is the relationship proportional? Explain using ratios and write a rule.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. A table has (x,y) pairs $[(1, 13), (2, 22), (4, 40)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 7), (2, 14), (4, 28)]$. Is the relationship proportional? Explain using ratios and write a rule.

3. A table has (x,y) pairs $[(1, 15), (2, 23), (4, 39)]$. Is the relationship proportional? Explain using ratios and write a rule.

4. A table has (x,y) pairs $[(1, 5), (2, 10), (4, 20)]$. Is the relationship proportional? Explain using ratios and write a rule.



Task-card worked solutions

1. No; $y = 3x + 2$

Ratios y/x are 5, 4, $7/2$. They differ. The starting value is not zero.

2. Yes; $y = 4x$

Ratios y/x are 4, 4, 4. They are equal, so the constant of proportionality is 4.

3. No; $y = 3x + 2$

Ratios y/x are 5, 4, $7/2$. They differ. The starting value is not zero.

4. Yes; $y = 9x$

Ratios y/x are 9, 9, 9. They are equal, so the constant of proportionality is 9.

5. No; $y = 7x + 5$

Ratios y/x are 12, $19/2$, $33/4$. They differ. The starting value is not zero.

6. Yes; $y = 5x$

Ratios y/x are 5, 5, 5. They are equal, so the constant of proportionality is 5.

Task-card worked solutions

7. No; $y = 4x + 5$

Ratios y/x are 9, $13/2$, $21/4$. They differ. The starting value is not zero.

8. Yes; $y = 2x$

Ratios y/x are 2, 2, 2. They are equal, so the constant of proportionality is 2.

9. No; $y = 2x + 6$

Ratios y/x are 8, 5, $7/2$. They differ. The starting value is not zero.

10. Yes; $y = 4x$

Ratios y/x are 4, 4, 4. They are equal, so the constant of proportionality is 4.

11. No; $y = 7x + 6$

Ratios y/x are 13, 10, $17/2$. They differ. The starting value is not zero.

12. Yes; $y = 6x$

Ratios y/x are 6, 6, 6. They are equal, so the constant of proportionality is 6.

Task-card worked solutions

13. No; $y = 5x + 6$

Ratios y/x are 11, 8, $13/2$. They differ. The starting value is not zero.

14. Yes; $y = 3x$

Ratios y/x are 3, 3, 3. They are equal, so the constant of proportionality is 3.

15. No; $y = 3x + 7$

Ratios y/x are 10, $13/2$, $19/4$. They differ. The starting value is not zero.

16. Yes; $y = 4x$

Ratios y/x are 4, 4, 4. They are equal, so the constant of proportionality is 4.

Warm-up worked solutions

1. No; $y = 8x + 2$

Ratios y/x are 10, 9, $17/2$. They differ. The starting value is not zero.

2. Yes; $y = 5x$

Ratios y/x are 5, 5, 5. They are equal, so the constant of proportionality is 5.

3. No; $y = 3x + 6$

Ratios y/x are 9, 6, $9/2$. They differ. The starting value is not zero.

4. Yes; $y = 7x$

Ratios y/x are 7, 7, 7. They are equal, so the constant of proportionality is 7.

5. No; $y = 3x + 5$

Ratios y/x are 8, $11/2$, $17/4$. They differ. The starting value is not zero.

6. Yes; $y = 2x$

Ratios y/x are 2, 2, 2. They are equal, so the constant of proportionality is 2.

Warm-up worked solutions

7. No; $y = 3x + 7$

Ratios y/x are 10, $13/2$, $19/4$. They differ. The starting value is not zero.

8. Yes; $y = 3x$

Ratios y/x are 3, 3, 3. They are equal, so the constant of proportionality is 3.

9. No; $y = 6x + 7$

Ratios y/x are 13, $19/2$, $31/4$. They differ. The starting value is not zero.

10. Yes; $y = 2x$

Ratios y/x are 2, 2, 2. They are equal, so the constant of proportionality is 2.

Classwork worked solutions

1. No; $y = 9x + 4$

Ratios y/x are 13, 11, 10. They differ. The starting value is not zero.

2. Yes; $y = 2x$

Ratios y/x are 2, 2, 2. They are equal, so the constant of proportionality is 2.

3. No; $y = 5x + 5$

Ratios y/x are 10, $15/2$, $25/4$. They differ. The starting value is not zero.

4. Yes; $y = 8x$

Ratios y/x are 8, 8, 8. They are equal, so the constant of proportionality is 8.

5. No; $y = 4x + 2$

Ratios y/x are 6, 5, $9/2$. They differ. The starting value is not zero.

6. Yes; $y = 6x$

Ratios y/x are 6, 6, 6. They are equal, so the constant of proportionality is 6.

Exit-ticket worked solutions

1. No; $y = 9x + 4$

Ratios y/x are 13, 11, 10. They differ. The starting value is not zero.

2. Yes; $y = 7x$

Ratios y/x are 7, 7, 7. They are equal, so the constant of proportionality is 7.

3. No; $y = 8x + 7$

Ratios y/x are 15, $23/2$, $39/4$. They differ. The starting value is not zero.

4. Yes; $y = 5x$

Ratios y/x are 5, 5, 5. They are equal, so the constant of proportionality is 5.

Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A table has (x,y) pairs $[(1, 6), (2, 8), (4, 12)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 2), (2, 4), (4, 8)]$. Is the relationship proportional? Explain using ratios and write a rule.

3. A table has (x,y) pairs $[(1, 14), (2, 21), (4, 35)]$. Is the relationship proportional? Explain using ratios and write a rule.

4. A table has (x,y) pairs $[(1, 6), (2, 12), (4, 24)]$. Is the relationship proportional? Explain using ratios and write a rule.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A table has (x,y) pairs $[(1, 8), (2, 13), (4, 23)]$. Is the relationship proportional? Explain using ratios and write a rule.

6. A table has (x,y) pairs $[(1, 3), (2, 6), (4, 12)]$. Is the relationship proportional? Explain using ratios and write a rule.

7. A table has (x,y) pairs $[(1, 14), (2, 21), (4, 35)]$. Is the relationship proportional? Explain using ratios and write a rule.

8. A table has (x,y) pairs $[(1, 7), (2, 14), (4, 28)]$. Is the relationship proportional? Explain using ratios and write a rule.



Fall • Worked solutions

1. No; $y = 2x + 4$

Ratios y/x are 6, 4, 3. They differ. The starting value is not zero.

2. Yes; $y = 2x$

Ratios y/x are 2, 2, 2. They are equal, so the constant of proportionality is 2.

3. No; $y = 7x + 7$

Ratios y/x are 14, $21/2$, $35/4$. They differ. The starting value is not zero.

4. Yes; $y = 6x$

Ratios y/x are 6, 6, 6. They are equal, so the constant of proportionality is 6.

Fall • Worked solutions

5. No; $y = 5x + 3$

Ratios y/x are 8, $13/2$, $23/4$. They differ. The starting value is not zero.

6. Yes; $y = 3x$

Ratios y/x are 3, 3, 3. They are equal, so the constant of proportionality is 3.

7. No; $y = 7x + 7$

Ratios y/x are 14, $21/2$, $35/4$. They differ. The starting value is not zero.

8. Yes; $y = 7x$

Ratios y/x are 7, 7, 7. They are equal, so the constant of proportionality is 7.

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A table has (x,y) pairs $[(1, 10), (2, 13), (4, 19)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 9), (2, 18), (4, 36)]$. Is the relationship proportional? Explain using ratios and write a rule.

3. A table has (x,y) pairs $[(1, 10), (2, 17), (4, 31)]$. Is the relationship proportional? Explain using ratios and write a rule.

4. A table has (x,y) pairs $[(1, 5), (2, 10), (4, 20)]$. Is the relationship proportional? Explain using ratios and write a rule.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A table has (x,y) pairs $[(1, 13), (2, 21), (4, 37)]$. Is the relationship proportional? Explain using ratios and write a rule.

6. A table has (x,y) pairs $[(1, 5), (2, 10), (4, 20)]$. Is the relationship proportional? Explain using ratios and write a rule.

7. A table has (x,y) pairs $[(1, 7), (2, 9), (4, 13)]$. Is the relationship proportional? Explain using ratios and write a rule.

8. A table has (x,y) pairs $[(1, 3), (2, 6), (4, 12)]$. Is the relationship proportional? Explain using ratios and write a rule.



Winter • Worked solutions

1. No; $y = 3x + 7$

Ratios y/x are 10, $13/2$, $19/4$. They differ. The starting value is not zero.

2. Yes; $y = 9x$

Ratios y/x are 9, 9, 9. They are equal, so the constant of proportionality is 9.

3. No; $y = 7x + 3$

Ratios y/x are 10, $17/2$, $31/4$. They differ. The starting value is not zero.

4. Yes; $y = 5x$

Ratios y/x are 5, 5, 5. They are equal, so the constant of proportionality is 5.

Winter • Worked solutions

5. No; $y = 8x + 5$

Ratios y/x are 13, $21/2$, $37/4$. They differ. The starting value is not zero.

6. Yes; $y = 5x$

Ratios y/x are 5, 5, 5. They are equal, so the constant of proportionality is 5.

7. No; $y = 2x + 5$

Ratios y/x are 7, $9/2$, $13/4$. They differ. The starting value is not zero.

8. Yes; $y = 3x$

Ratios y/x are 3, 3, 3. They are equal, so the constant of proportionality is 3.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A table has (x,y) pairs $[(1, 10), (2, 17), (4, 31)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 5), (2, 10), (4, 20)]$. Is the relationship proportional? Explain using ratios and write a rule.

3. A table has (x,y) pairs $[(1, 11), (2, 15), (4, 23)]$. Is the relationship proportional? Explain using ratios and write a rule.

4. A table has (x,y) pairs $[(1, 7), (2, 14), (4, 28)]$. Is the relationship proportional? Explain using ratios and write a rule.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A table has (x,y) pairs $[(1, 9), (2, 15), (4, 27)]$. Is the relationship proportional? Explain using ratios and write a rule.

6. A table has (x,y) pairs $[(1, 8), (2, 16), (4, 32)]$. Is the relationship proportional? Explain using ratios and write a rule.

7. A table has (x,y) pairs $[(1, 11), (2, 17), (4, 29)]$. Is the relationship proportional? Explain using ratios and write a rule.

8. A table has (x,y) pairs $[(1, 4), (2, 8), (4, 16)]$. Is the relationship proportional? Explain using ratios and write a rule.



Spring break • Worked solutions

1. No; $y = 7x + 3$

Ratios y/x are 10, $17/2$, $31/4$. They differ. The starting value is not zero.

2. Yes; $y = 5x$

Ratios y/x are 5, 5, 5. They are equal, so the constant of proportionality is 5.

3. No; $y = 4x + 7$

Ratios y/x are 11, $15/2$, $23/4$. They differ. The starting value is not zero.

4. Yes; $y = 7x$

Ratios y/x are 7, 7, 7. They are equal, so the constant of proportionality is 7.

Spring break • Worked solutions

5. No; $y = 6x + 3$

Ratios y/x are 9, $15/2$, $27/4$. They differ. The starting value is not zero.

6. Yes; $y = 8x$

Ratios y/x are 8, 8, 8. They are equal, so the constant of proportionality is 8.

7. No; $y = 6x + 5$

Ratios y/x are 11, $17/2$, $29/4$. They differ. The starting value is not zero.

8. Yes; $y = 4x$

Ratios y/x are 4, 4, 4. They are equal, so the constant of proportionality is 4.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A table has (x,y) pairs $[(1, 11), (2, 16), (4, 26)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 8), (2, 16), (4, 32)]$. Is the relationship proportional? Explain using ratios and write a rule.

3. A table has (x,y) pairs $[(1, 12), (2, 21), (4, 39)]$. Is the relationship proportional? Explain using ratios and write a rule.

4. A table has (x,y) pairs $[(1, 4), (2, 8), (4, 16)]$. Is the relationship proportional? Explain using ratios and write a rule.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A table has (x,y) pairs $[(1, 10), (2, 17), (4, 31)]$. Is the relationship proportional? Explain using ratios and write a rule.

6. A table has (x,y) pairs $[(1, 9), (2, 18), (4, 36)]$. Is the relationship proportional? Explain using ratios and write a rule.

7. A table has (x,y) pairs $[(1, 10), (2, 17), (4, 31)]$. Is the relationship proportional? Explain using ratios and write a rule.

8. A table has (x,y) pairs $[(1, 4), (2, 8), (4, 16)]$. Is the relationship proportional? Explain using ratios and write a rule.



Summer • Worked solutions

1. No; $y = 5x + 6$

Ratios y/x are 11, 8, $13/2$. They differ. The starting value is not zero.

2. Yes; $y = 8x$

Ratios y/x are 8, 8, 8. They are equal, so the constant of proportionality is 8.

3. No; $y = 9x + 3$

Ratios y/x are 12, $21/2$, $39/4$. They differ. The starting value is not zero.

4. Yes; $y = 4x$

Ratios y/x are 4, 4, 4. They are equal, so the constant of proportionality is 4.

Summer • Worked solutions

5. No; $y = 7x + 3$

Ratios y/x are 10, $17/2$, $31/4$. They differ. The starting value is not zero.

6. Yes; $y = 9x$

Ratios y/x are 9, 9, 9. They are equal, so the constant of proportionality is 9.

7. No; $y = 7x + 3$

Ratios y/x are 10, $17/2$, $31/4$. They differ. The starting value is not zero.

8. Yes; $y = 4x$

Ratios y/x are 4, 4, 4. They are equal, so the constant of proportionality is 4.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Identify and represent proportional relationships. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Compare y/x at nonzero x values. A constant ratio gives $y = kx$ and a graph through the origin.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A table has (x,y) pairs $[(1, 11), (2, 17), (4, 29)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 5), (2, 10), (4, 20)]$. Is the relationship proportional? Explain using ratios and write a rule.

3. A table has (x,y) pairs $[(1, 8), (2, 12), (4, 20)]$. Is the relationship proportional? Explain using ratios and write a rule.

4. A table has (x,y) pairs $[(1, 4), (2, 8), (4, 16)]$. Is the relationship proportional? Explain using ratios and write a rule.



Substitute lesson • Worked solutions

1. No; $y = 6x + 5$

Ratios y/x are 11, $17/2$, $29/4$. They differ. The starting value is not zero.

2. Yes; $y = 5x$

Ratios y/x are 5, 5, 5. They are equal, so the constant of proportionality is 5.

3. No; $y = 4x + 4$

Ratios y/x are 8, 6, 5. They differ. The starting value is not zero.

4. Yes; $y = 4x$

Ratios y/x are 4, 4, 4. They are equal, so the constant of proportionality is 4.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Identify and represent proportional relationships. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Compare y/x at nonzero x values. A constant ratio gives $y = kx$ and a graph through the origin.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A table has (x,y) pairs $[(1, 12), (2, 20), (4, 36)]$. Is the relationship proportional? Explain using ratios and write a rule.

2. A table has (x,y) pairs $[(1, 5), (2, 10), (4, 20)]$. Is the relationship proportional? Explain using ratios and write a rule.

3. A table has (x,y) pairs $[(1, 5), (2, 7), (4, 11)]$. Is the relationship proportional? Explain using ratios and write a rule.

4. A table has (x,y) pairs $[(1, 8), (2, 16), (4, 32)]$. Is the relationship proportional? Explain using ratios and write a rule.



Emergency substitute lesson • Worked solutions

1. No; $y = 8x + 4$

Ratios y/x are 12, 10, 9. They differ. The starting value is not zero.

2. Yes; $y = 5x$

Ratios y/x are 5, 5, 5. They are equal, so the constant of proportionality is 5.

3. No; $y = 2x + 3$

Ratios y/x are 5, $7/2$, $11/4$. They differ. The starting value is not zero.

4. Yes; $y = 8x$

Ratios y/x are 8, 8, 8. They are equal, so the constant of proportionality is 8.

A completed lesson-plan example

Level and focus

Grade 7; NC.7.RP.2; Identify and represent proportional relationships.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.7.RP.2 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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